

Encryption Algorithm Optimization for Practical Cyber

Security Applications

Final Year Project Proposal

Submitted by

Rafia Rehman

F22BNSE1M02003

Aliza Yasin

F22BINSE1M02009

Supervised by

Ms. Rabia Kamran

BS Cyber Security

Session 2022-2026

Department of Information Security

Faculty of Computing



The Islamia University of Bahawalpur

2026

Abstract

With the rapid growth of digital systems, secure and efficient data protection has become a critical requirement in cyber security. Encryption plays a vital role in protecting sensitive information from unauthorized access; however, strong encryption algorithms often introduce computational overhead, which can affect system performance. This project focuses on the optimization of an existing encryption algorithm to improve its practical usability without compromising security.

The proposed project aims to analyze the Advanced Encryption Standard (AES), a widely adopted symmetric encryption algorithm, and develop an optimized approach that enhances encryption speed while maintaining the same level of confidentiality. The optimization is achieved by employing efficient encryption modes and minimizing unnecessary computational operations during the encryption process. The system will be implemented using Python, allowing secure file-based encryption and performance comparison between the standard and optimized approaches.

The effectiveness of the proposed solution will be evaluated by encrypting files of different sizes and measuring execution time. The results will demonstrate how the optimized encryption approach reduces processing time and improves efficiency, making it suitable for real-world cyber security applications such as secure file storage and data transmission. This project provides a practical understanding of encryption performance issues and highlights the importance of balancing security and computational efficiency in modern cyber security systems.

Keywords: Cyber Security, Encryption, AES, Data Security, Performance Optimization.

Table of Contents

Abstract	1
Table of Contents	3
List of Figures	4
List of Tables	5
Chapter 1 Introduction	6
1.1 Background/Overview	6
1.2 Problem Statement	6
1.3 Aims and Objectives	6
1.4 Scope of the Project	7
Chapter 2 Related Work	8
Chapter 3 Methodology	9
3.1 Proposed Method	9
3.2 System Flow Diagram (Textual Representation).....	9
3.3 Algorithm Description.....	10
Chapter 4 Expected Outcomes	11
4.1 Tools and Technologies Required	11
4.1.1 Hardware	11
4.1.2 Software	11
4.2 Expected Results:	11
4.3 Solution Application Area	11
Chapter 5 Project Planning	12
5.1 Individual Tasks/Distribution of Work (mandatory if group project)	12
5.2 Gantt Chart	12
Appendix.....	13

List of Figures

Figure 1: Flow diagram optimization system.....	9
Figure 2: Flowchart of Standard and optimized AES Encryption Algorithm.....	13
Figure 3: Architecture diagram of the proposed encryption optimization system.....	14

List of Tables

Table 1.Distribution of task.....	12.
Table2. Gantt chart illustrating the Work plan.....	12

Chapter 1

Introduction

1.1 Background/Overview

Encryption is one of the fundamental components of information security and cyber security. It ensures confidentiality by converting plaintext data into unreadable ciphertext using cryptographic algorithms. Among various encryption standards, the Advanced Encryption Standard (AES) is one of the most widely used symmetric encryption algorithms due to its strong security and standardization by NIST. However, the use of encryption in real-time and resource-constrained environments may result in performance overhead. Therefore, optimizing encryption algorithms for practical use is an important research area in cyber security.

1.2 Problem Statement

Although AES provides strong security, its computational cost may impact system performance when encrypting large volumes of data or operating in real-time applications. This performance overhead can limit its usability in practical cyber security systems. The problem addressed in this project is how to optimize AES encryption to reduce execution time while maintaining its security properties.

1.3 Aims and Objectives

The main aim of this project is to optimize an encryption algorithm for improved performance in practical cyber security applications.

Objectives:

- To study the working of AES encryption algorithm.
- To identify performance limitations in standard AES implementation.
- To develop an optimized encryption approach using AES.
- To compare the performance of standard and optimized encryption methods.

1.4 Scope of the Project

The scope of this project is limited to symmetric encryption using AES. The project focuses on file-based encryption and performance analysis using Python. Advanced cryptanalysis and hardware-level optimizations are not included in the scope of this work.

Chapter 2

Related Work

Several studies have been conducted on encryption optimization to improve performance in real-time and resource-limited systems. Previous research has focused on lightweight encryption algorithms, optimized encryption modes, and efficient key management techniques. While many approaches introduce new encryption schemes, this project differs by optimizing a well-established and standardized algorithm (AES) to maintain compatibility and security while improving performance..

Chapter 3

Methodology

3.1 Proposed Method

The proposed methodology is based on implementing and comparing two encryption approaches: standard AES encryption and optimized AES encryption. The optimization focuses on improving execution speed while maintaining the same security level. The system processes input files, encrypts their data, measures performance, and compares results.

Step-by-step working:

1. Input file is selected from the system
2. File data is read using Python file handling
3. AES key and Initialization Vector (IV) are generated
4. Standard AES encryption is applied
5. Optimized AES encryption is applied
6. Encryption time is measured for both methods
7. Results are compared and analyzed

3.2 System Flow Diagram (Textual Representation)

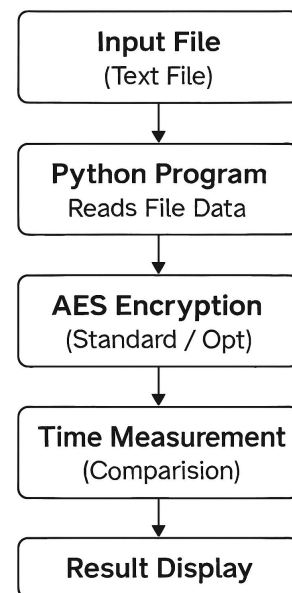


Figure 1. Flow diagram of the proposed encryption optimization system

3.3 Algorithm Description

Algorithm 1: Standard AES Encryption

1. Start
2. Read input file
3. Generate AES key and IV
4. Apply AES encryption
5. Measure execution time
6. Display results
7. End

Algorithm 2: Optimized AES Encryption

1. Start
2. Read input file
3. Generate AES key and IV
4. Apply optimized AES encryption mode
5. Measure execution time
6. Display results
7. End

Chapter 4

Expected Outcomes

The expected outcome of this project is an optimized encryption system that demonstrates improved performance compared to standard AES encryption while maintaining security.

4.1 Tools and Technologies Required

4.1.1 Hardware

- *Development PC with Windows 10 or later*
- Minimum 4GB RAM

4.1.2 Software

- Python 3.x
- Cryptography library
- Microsoft Word (documentation)

4.2 Expected Results

The optimized encryption method is expected to show reduced execution time during encryption compared to standard AES, especially for small and medium-sized files.

4.3 Solution Application Area

The proposed solution can be applied in secure file storage systems, data transmission systems, and real-time cyber security applications where performance and security are both critical.

Chapter 5

Project Planning

5.1 Individual Tasks/Distribution of Work

Table 1 Distribution of task

Task	Group Member 1 Rafia Rehman	Group Member 2 Aliza Yasin
Literature Review	○	●
AES Implementation	○	●
Optimization Design	●	○
Testing & Evaluation	●	○
Documentation		●

5.2 Gantt Chart

Table 2 Gantt chart illustrating the Work plan

Task	Feb	Marh	Aprl	May
– Proposal Submission				
– Literature Review				
– Start implementation				
– Testing & Evaluation				
– Documentation				
– Proposal Submission				

Appendix A: Algorithm Flowchart Diagrams

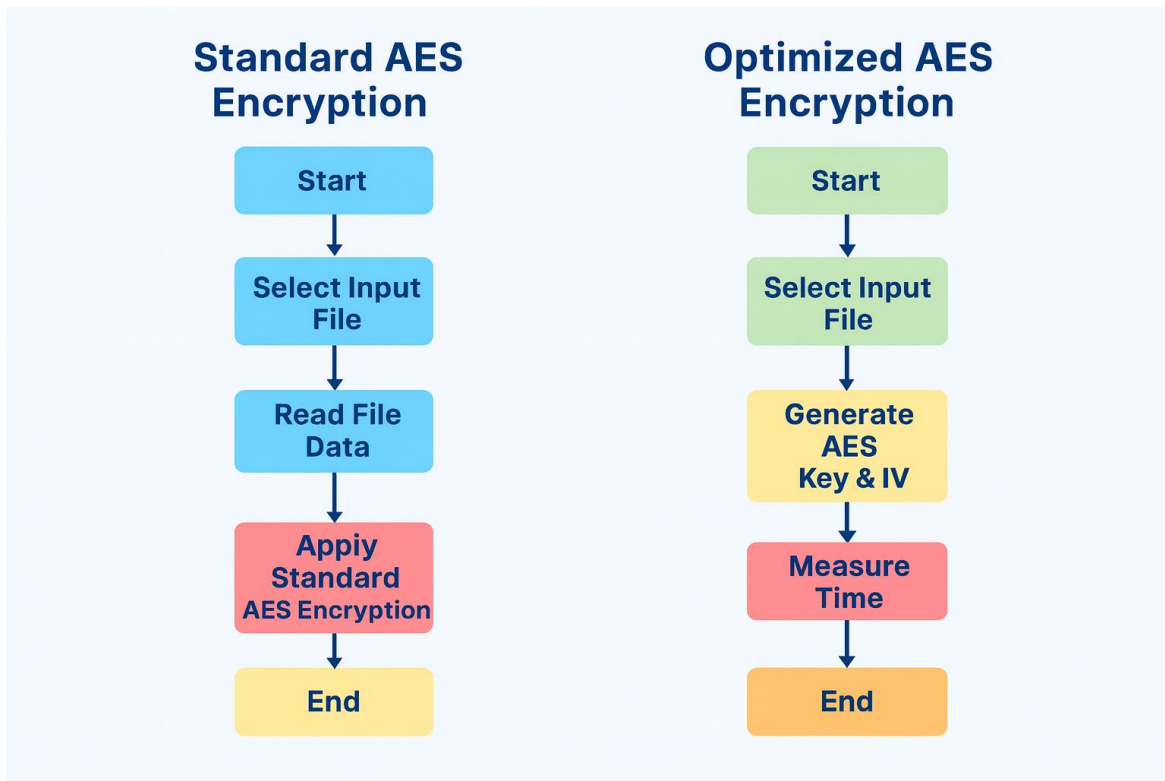


Figure 2: Flowchart of Standard and optimized AES Encryption Algorithm

Appendix B: System Architecture Diagram

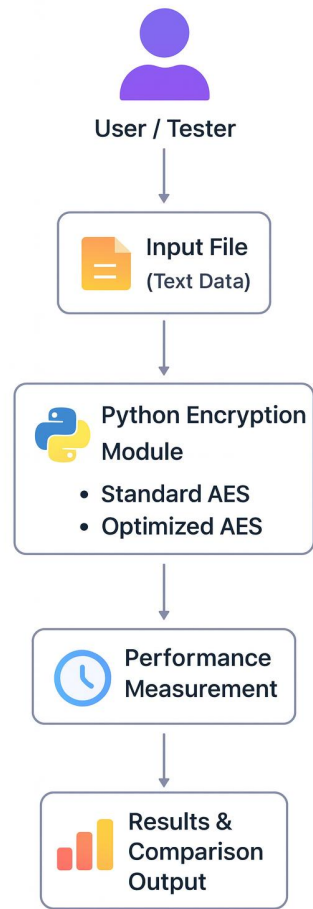


Figure 3: Architecture diagram of the proposed encryption optimization system